

Arwin Karir

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EDUCATION

University of Toronto

BASc in Computer Engineering (PEY Co-op)

Sept. 2025 – May 2030

Toronto, ON

EXPERIENCE

Software Engineer

May 2026 – Present

SellStatic

Toronto, ON

- Built end-to-end customer onboarding system that imports business websites, extracts brand assets and color palettes from SVG and raster logos, automatically configures ad-generation workflows, and transactionally persists user and brand data through a C# API, supported by 10+ targeted tests
- Architected an asynchronous AI-video delivery pipeline spanning React, ASP.NET Core, Azure Blob Storage, MongoDB, and AWS CDK; used SNS, SQS with a dead-letter queue, Lambda, and SES to persist completed renders and notify users after they safely leave the application
- Shipped major upgrades to SellStatic's React/TypeScript video editor, including layered timelines, rich text and font controls, multi-track audio and volume editing, save-state protection, media handling, and streamlined export workflows
- Transformed the Fabric.js image editor into a workspace with canvas zoom and navigation, layer management, media uploads, AI-generated asset libraries, template-wide color editing, and undo/redo support; added five Playwright end-to-end scenarios covering critical editor workflows.

Artificial Intelligence Intern

Jun. 2023 – Aug. 2023

Amazon Work Experience Program

Remote

- Built a supervised machine learning workflow on environmental datasets for forest-cover classification, structuring experiments for reproducible training, evaluation, and model comparison
- Owned Python-based preprocessing and feature engineering with pandas, NumPy, and scikit-learn to clean inputs, standardize features, and improve training signal quality
- Benchmarked k-NN, logistic regression, and neural network models, reporting trade-offs across accuracy, interpretability, and computational cost to guide model selection
- Used AWS development workflows with Amazon SageMaker, Amazon S3, and Amazon EC2, collaborating on a 4-person team to present metrics-driven results and next steps to AWS engineers

PROJECTS

LumiSense AI | *PyTorch, ONNX, OpenCV, Raspberry Pi*

Jan. 2026 – Present

- Implemented a stakeholder-facing edge AI solution for the UofT Sustainability Office to estimate floor-plane illuminance and support campus building lighting performance analysis
- Developed a multitask vision model to estimate floor-plane illuminance by predicting a dense lux map, summary statistics, and point-wise lux queries under/between fixtures
- Implemented an encoder-decoder architecture with a ResNet-style backbone, U-Net-style upsampling, and task-specific heads for segmentation, lux regression, and uncertainty/quality estimation
- Added deployment reliability features including luminance-based frame-quality gating, debug logging, and snapshots while generating outputs for UofT Sustainability Office lighting performance reporting

CUDA Optimization of Fused Causal Softmax | *CUDA, C++, PyTorch, pybind11, Python*

July 2026 – Present

- Developed a PyTorch CUDA extension for fused FP32 causal scaled softmax, combining scaling, causal masking, numerically stable exponentiation, reduction, and normalization into a block-parallel softmax kernel.
- Outperformed equivalent PyTorch eager softmax by $1.40\text{--}3.94\times$ on an NVIDIA Tesla T4 across sequence lengths 128–2048 and beat steady-state torch.compile at six of seven tested shapes
- Integrated the custom operator into explicit transformer attention, delivering $1.54\text{--}2.31\times$ end-to-end speedup over explicit PyTorch attention while analyzing performance limits against production PyTorch SDPA
- Validated 88/88 numerical and edge-shape cases with maximum absolute error of 3.58×10^{-7} , and built commit-tagged benchmarking, profiling, provenance, and reproducibility workflows using CUDA events, PyTorch Profiler, Nsight Compute, and Google Colab

TECHNICAL SKILLS

Languages: C#, Python, C/C++, TypeScript, JavaScript, SQL

Frameworks/Libraries: React, Next.js, ASP.NET Core, PyTorch, ONNX Runtime, OpenCV, Playwright

Cloud/Data: AWS Lambda, SQS, SNS, SES, S3, CDK, SageMaker; Azure Blob Storage; MongoDB; PostgreSQL

Systems/Tools: CUDA, Nsight Compute, PyTorch Profiler, Raspberry Pi, Linux, Git